Dr. Benjamin Lee Davis — Curriculum Vitæ

Websites — Personal: https://bendavis007.github.io • Work: http://bit.ly/benCAP3 • Old Work: https://astronomy.swin.edu.au/staff/benjamindavis.html
Mailing Addresses — Home: 6970 SE Kiwi Ln, Riverton, KS 66770, USA • Office: 509½ S Main St, Galena, KS 66739, USA
Email Addresses — ben.davis@nyu.edu • benjamindavis@astro.swin.edu.au • bendavis007@hotmail.com • benjaminleedavis007@gmail.com
Profiles — https://www.linkedin.com/in/bendavis007 • https://www.researchgate.net/profile/Benjamin_Davis9 • https://orcid.org/0000-0002-4306-5950
Telephone Numbers — Home: +1 (620) 848-3981 • Mobile: +1 (620) 215-1547

EDUCATION

University of Arkansas, Fayetteville, Arkansas, USA — Aug 2008 – May 2015

Doctor of Philosophy (Ph.D.) in Space & Planetary Sciences
Thesis: Logarithmic Spiral Arm Pitch Angle of Spiral Galaxies: Measurement and Relationship to Galactic Structure and Nuclear Supermassive Black Hole Mass
Advisor: Dr. Julia Kennefick
Research areas: Astrophysics, extragalactic astronomy, structure & dynamics of disk galaxies.
Cumulative GPA: 4.0000 / 4.0000 (117 credit hours)

- News Articles
  - The Galena Sentinel Times: "Out of this World"
  - The Galena Sentinel Times: "Davis Receives His Doctorate of Philosophy"

Pittsburg State University, Pittsburg, Kansas, USA — Aug 2003 – May 2008

- Bachelor of Science (B.S.) in Mathematics
- Bachelor of Science (B.S.) in Physics
- Minor in Music

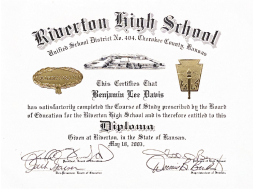
Magna Cum Laude
Cumulative GPA: 3.9397 / 4.0000 (199 credit hours)

Riverton High School, Riverton, Kansas, USA — Aug 1999 – May 2003

Valedictorian & Senior Class President
Cumulative GPA: 4.0000 / 4.0000

University of Arkansas
To all to whom these presents may come
Greeting
Be it known that
Benjamin Lee Davis
having completed the prescribed course of study to the satisfaction of the Board of Trustees of the University of Arkansas submitted to the Degree of
Doctor of Philosophy
and is declared entitled to all rights and privileges which pertain to that degree here or elsewhere. In witness whereof and in evidence that all requirements for the degree have been fulfilled, this diploma is granted.
Dated at the University in the City of Fayetteville, this third day of May in the year of our Lord two thousand and fifteen and of the University the one hundred and forty-fourth year.

Pittsburg State University
Benjamin L. Davis
has begun
Bachelor of Science
Bachelor of Arts
with all the Honors and Distinction belonging to this Degree in consideration of the satisfactory completion of the Course of Study prescribed by the Faculty of the University.
Given at Pittsburg, Kansas, on the seventeenth day of May, in the year of our Lord two thousand and eight.


Riverton High School
Unified School District No. 404, Cherokee County, Kansas
This Certifies That
Benjamin Lee Davis
has satisfactorily completed the Course of Study prescribed by the Board of Education for the Riverton High School and is therefore entitled to this
Diploma
Given at Riverton, in the State of Kansas,
May 16, 2003.

WORK EXPERIENCE

New York University Abu Dhabi — Sep 2020 – Aug 2023
Center for Astro, Particle, and Planetary Physics (CAP^3)
CAP^3 Fellow, Research Associate
Supervisor: Prof. Andrea Valerio Macciò

- News Articles
  - The Galena Sentinel Times: "In Our Midst – Fast Forward"

Swinburne University of Technology — Aug 2016 – Jul 2020
Centre for Astrophysics and Supercomputing
Postdoctoral Research Fellow in Galaxy Structure
Project: Reducing and analyzing galaxy images via bulge/bar/disc decompositions. I am responsible for modeling the radial distribution of optical and/or near-infrared stellar light in nearby (<200 Mpc) galaxies with an emphasis on discoveries and publications related to the (central black hole)-(host bulge) connection and/or compact massive spheroids and/or galaxies with partially depleted cores or additional nuclear components.
Supervisor: Prof. Alister Graham
Research areas: Astrophysics, extragalactic astronomy, structure & dynamics of disk galaxies.

- Additional service:
  - Accredited Supervisor for Masters and Ph.D. Students.
    - * Associate Supervisor for Ph.D. Student Nandini Sahu.
    - * Associate Supervisor for Ph.D. Student Suei-Hei (Dexter) Hon.
  - Served on the Ph.D. Student Review Panels for Ellert van der Velden, Jonah Gannon, Liyualem Tilahun, and Arianna Dolfi.

Arkansas Tech University, Department of Physical Sciences — Aug 2015 – May 2016
Visiting Assistant Professor of Physics

University of Arkansas, Department of Physics — Aug 2015 – Dec 2015
Visiting Assistant Professor of Physics

RESEARCH EXPERIENCE

LISA Consortium — May 2020 –
Member

ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav) — Feb 2018 –
Affiliate

Arkansas Galaxy Evolution Survey, University of Arkansas — May 2015 – Aug 2015
Postdoctoral Researcher
Project: Measuring the Number of Arms in Spiral Galaxies and Relating that to Predictions of Density Wave and Swing-Amplification Theories
Supervisors: Drs. Julia and Daniel Kennefick
Research areas: Astrophysics, extragalactic astronomy, structure & dynamics of disk galaxies.

Arkansas Galaxy Evolution Survey, University of Arkansas — Jan 2009 – May 2015
Graduate Research Assistant
Project: Logarithmic Spiral Arm Pitch Angle of Spiral Galaxies: Measurement and Relationship to Galactic Structure and Nuclear Supermassive Black Hole Mass
Supervisors: Drs. Julia and Daniel Kennefick
Research areas: Astrophysics, extragalactic astronomy, structure & dynamics of disk galaxies.

Arkansas Center for Space and Planetary Sciences, University of Arkansas — Aug 2008 – May 2015
Graduate Student
Research areas: Martian surface chemistry and exoplanet demographics.

Cluster Lensing and Supernova Survey with Hubble, Jet Propulsion Laboratory
Visiting Student Research Program Intern — Jun 2011 – Aug 2011
Project: Spectral Energy Distribution Fitting to Galaxy Cluster Abell 383
Supervisor: Dr. Leonidas Moustakas
Research areas: Dark matter, galaxy clusters, Abell 383, high-redshift galaxies, gravitational lensing, spectral energy distributions.

PEER-REVIEWED PUBLICATIONS

22) Davis, B. L. & Graham, A. W., "Refining the mass estimate for the intermediate-mass black hole candidate in NGC 3319," submitted to the *Publications of the Astronomical Society of Australia*, March 9, 2021.

21) Graham, A, W., Soria, R., Davis, B. L., & Swartz, D. A., "Spiral galaxies of the Virgo cluster: possible black hole seeding/feeding of the bulgeless spiral galaxy NGC 4424 via an infalling star cluster," submitted to the *Monthly Notices of the Royal Astronomical Society*, March 3, 2021.

20) Graham, A, W., Soria, R., Davis, B. L., Kolehmainen, M., Maccarone, T., Miller-Jones, J., Motch, C., & Swartz, D. A., "Spiral galaxies of the Virgo cluster: Eleven more intermediate-mass black hole candidates with an associated X-ray point-source," submitted to the *Monthly Notices of the Royal Astronomical Society*, December 16, 2020.

19) Abdeen, S., Kennefick, D., Kennefick, J., Eufrasio, R., Davis, B. L., Miller, R., Shields, D. W., Monson, E. B., Bassett, C., & O'Mara, H., "Evidence in favour of density wave theory through star formation history maps and spatially resolved stellar clusters," submitted to the *Monthly Notices of the Royal Astronomical Society*, October 24, 2020.

18) Sahu, N., Graham, A. W., & Davis, B. L., "Defining the (Black Hole)–Spheroid Connection with the Discovery of Morphology-dependent Substructure in the $M_{\rm BH}$–$n_{\rm sph}$ and $M_{\rm BH}$–$R_{\rm e,sph}$ Diagrams: New Tests for Advanced Theories and Realistic Simulations," *The Astrophysical Journal, 903(2), 97*, November 6, 2020.

17) Abdeen, S., Kennefick, D., Kennefick, J., Miller, R., Shields, D. W., Monson, E., & Davis, B. L., "Determining the Co-Rotation Radii of Spiral Galaxies Using Spiral Arm Pitch Angle Measurements at Multiple Wavelengths," *Monthly Notices of the Royal Astronomical Society, 496, 1610*, June 8, 2020.

16) Sahu, N., Graham, A. W., & Davis, B. L., "Revealing Hidden Substructures in the $M_{\rm BH}$–σ Diagram, and Refining the Bend in the L–σ Relation," *The Astrophysical Journal, 887, 10*, December 10, 2019.

15) Davis, B. L., Graham, A. W., & Combes, F., "A Consistent Set of Empirical Scaling Relations for Spiral Galaxies: The $(v_{\rm max},\ M_{\rm DM})$–$(\sigma_0,\ M_{\rm BH},\ \phi)$ Relations," *The Astrophysical Journal, 877, 64*, May 24, 2019.

14) Sahu, N., Graham, A. W., & Davis, B. L., "Black Hole Mass Scaling Relations for Early-Type Galaxies. I. $M_{\rm BH}$–$M_{*,\rm sph}$ and $M_{\rm BH}$–$M_{*,\rm gal}$," *The Astrophysical Journal, 876, 155*, May 15, 2019.

13) Miller, R., Kennefick, D., Kennefick, J., Shameer Abdeen, M, Monson, E., Eufrasio, R. T., Shields, D. W., & Davis, B. L., "Investigating the Origins of Spiral Structure in Disk Galaxies Through a Multi-wavelength Study," *The Astrophysical Journal, 874, 177*, April 5, 2019.

- News Article Coverage
  - University of Arkansas News: "Research on Disk Galaxies Sheds Light on Movement of Stars"

12) Graham, A. W., Soria, R., & Davis, B. L., "Expected intermediate mass black holes in the Virgo cluster. II. Late-type galaxies," *Monthly Notices of the Royal Astronomical Society, 484, 814*, March 21, 2019.

11) Davis, B. L., Graham, A. W., & Cameron, E., "Black Hole Mass Scaling Relations for Spiral Galaxies. I. $M_{\rm BH}$–$M_{*,\rm sph}$," *The Astrophysical Journal, 873, 85*, March 6, 2019.

10) Davis, B. L., Graham, A. W., & Cameron, E., "Black Hole Mass Scaling Relations for Spiral Galaxies. II. $M_{\rm BH}$–$M_{*,\rm tot}$ and $M_{\rm BH}$–$M_{*,\rm disk}$," *The Astrophysical Journal, 869, 113*, December 17, 2018.

9) Davis, B. L., Graham, A. W., & Seigar, M. S., "Updating the (Supermassive Black Hole Mass) – (Spiral Arm Pitch Angle) Relation: A Strong Correlation for Galaxies with Pseudobulges," *Monthly Notices of the Royal Astronomical Society, 471, 2187*, October 21, 2017.

- Jul 2017, Swinburne Media Release: "Spiral Arms Allow School Children to Weigh Black Holes"
  - UK Version (MNRAS)
- English Media: IFL Science, Gizmodo, Science Daily, PHYS.ORG, AZO Quantum, COSMOS Magazine, Space Daily, ineffableisland.com, MS-CHAT.COM, Across the Universe, AlphaGalileo, Health Medicine Network, EurekAlert, sciencesprings, & Astro Watch.
- Non-English Media: Greek1, Greek2, Greek3, Italian1, Italian2, Italian3, German, Dutch1, Dutch2, Dutch3, Czech, Polish, Spanish, Portuguese, Vietnamese1, Vietnamese2, Vietnamese3, Vietnamese4, Vietnamese5, Indonesian, & Japanese.
- Radio:
  - Duluth Public Radio KUMD 103.3 FM
  - Space Nuts with Dr Fred Watson & Andrew Dunkley (1:07 - 10:10)
  - SpaceTime with Stuart Gary (7:31 - 18:14)

8) Koliopanos, F., Ciambur, B. C., Graham, A. W., Webb, N. A., Coriat, M., Mutlu-Pakdil, B., Davis, B. L., Godet, O., Barret, D., & Seigar, M. S., "Searching for Intermediate Mass Black Holes in Dwarf Galaxies with Low Luminosity AGN: A Multiple-method Approach," *Astronomy & Astrophysics, Volume 601, A20*, Dec 2016.

7) Mutlu-Pakdil, B., Seigar, M. S., & Davis, B. L., "The Local Black Hole Mass Function Derived from the $M_{BH} - P$ and the $M_{BH} - n$ Relations," *The Astrophysical Journal, 830, 117*, Oct 2016.

6) Pour-Imani, H., Kennefick, D., Kennefick, J., Davis, B. L., Shields, D. W., & Shameer Abdeen, M., "Strong Evidence for the Density-wave Theory of Spiral Structure in Disk Galaxies," in *The Astrophysical Journal Letters, 827, L2*, Aug 2016.

- News Article Coverage
  - PHYS.ORG: "Astrophysicists Discover Mechanism for Spiral-Arm Formation in Disk Galaxies"
  - University of Arkansas News: "U of A Astrophysicists Discover Mechanism for Spiral-Arm Formation in Disk Galaxies"
  - Sky & Telescope: "Why Galaxies Have Spiral Arms"

5) Davis, B. L., Kennefick, D., Kennefick, J., Westfall, K. B., Shields, D. W., Flatman, R., Hartley, M. T., Berrier, J. C., Martinsson, T. P. K., & Swaters, R. A., "A Fundamental Plane of Spiral Structure in Disk Galaxies," *The Astrophysical Journal Letters, 802, L13*, Mar 2015.

4) Seigar, M. S., Davis, B. L., Berrier, J. C., & Kennefick, D., "Constraining Dark Matter Halo Profiles and Galaxy Formation Models Using Spiral Arm Morphology. II. Dark and Stellar Mass Concentrations from 13 Nearby Face-On Galaxies," *The Astrophysical Journal, 795, 90*, Nov 2014.

3) Davis, B. L., Berrier, J. C., Johns, L., Shields, D. W., Hartley, M. T., Kennefick, D., Kennefick, J., Seigar, M. S., & Lacy, C. H. S., "The Black Hole Mass Function Derived from Local Spiral Galaxies," *The Astrophysical Journal, 789, 124*, Jul 2014.

2) Berrier, J. C., Davis, B. L., Kennefick, D. Kennefick, J., Seigar, M. S., Barrows, R. S., Hartley, M. T., Shields, D. W., Bentz, M. C., & Lacy, C. H. S., "Further Evidence for a Supermassive Black Hole – Pitch Angle Relation," *The Astrophysical Journal, 769, 132*, Jun 2013.

1) Davis, B. L., Berrier, J. C., Shields, D. W., Kennefick, J., Kennefick, D., Seigar, M. S., Lacy, C. H. S., & Puerari I., "Measurement of Galactic Logarithmic Spiral Arm Pitch Angle Using Two-Dimensional Fast Fourier Transform Decomposition," *The Astrophysical Journal Supplement Series, 199, 33*, Apr 2012.

- Software - http://ascl.net/1608.015

If you are viewing this CV on paper, you may access a full list of my publications from my ORCID by visiting https://orcid.org/0000-0002-4306-5950 or scanning the QR code below.

NASA ADS Library Links

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All – https://ui.adsabs.harvard.edu/user/libraries/1IplwMbdTmunVHkWfUR6XA

MANUSCRIPTS

2) Fusco, M. S., Davis, B. L., Kennefick, D., Seigar, M. S., & Kennefick, J., "Pitch Angle Distribution Function of Local Spiral Galaxies and Its Role in Galactic Structure," (draft) Jul 2019.

1) Shields, D. W., Boe, B., Pfountz, C., Davis, B. L., Hartley, M., Pour Imani, H., Slade, Z., Kennefick, D., & Kennefick, J., "Spirality: A Novel Way to Measure Spiral Arm Pitch Angle," arXiv:1511.06365, Nov 2015.

- Software - http://ascl.net/1512.015

MAGAZINE ARTICLE

Davis, B. (2017). "How You Can Weigh Black Holes," *Australasian Science, (Volume 38 | Number 5), pp.14-16, Sept/Oct 2017.*

- Preprint - https://bendavis007.github.io/PDFs/Australasian_Science.pdf

PRESENTATIONS

26) Davis, B. L., "The Latest Black Hole Mass Scaling Relations: Revealing Morphology Dependence, Substructure, and Implications for the Coevolution of Black Holes with their Host Galaxies," *CAP^3 Seminar Series*, Oct 2020. [Seminar Talk]

25) Davis, B. L., "Title TBD," Seminar Speaker at the *International Centre for Radio Astronomy Research at the University of Western Australia*, Mar 2020 (postponed due to the COVID-19 pandemic). [Invited Talk]

24) Davis, B. L., "Substructure in Black Hole Scaling Diagrams and Implications for the Coevolution of Black Holes and Galaxies," *IAU Symposium 359: Galaxy Evolution and Feedback Across Different Environments (GALFEED)*, Mar 2020. [Poster Presentation]

- Poster
- Proceedings (Preprint)

23) Davis, B. L., "Black Hole Mass Scaling Relations," Seminar Speaker at the *Center for Astro, Particle, and Planetary Physics at New York University Abu Dhabi*, Feb 2020. [Invited Talk]

22) Davis, B. L., "Beethoven and the Voyager Golden Records," *Melbourne Chamber Orchestra's Beethoven Gala Dinner*, Oct 2019. [Invited Talk]

21) Davis, B. L. & Graham, A. W., "The Latest Black Hole Mass Scaling Relations," *Feedback and its Role in Galaxy Formation*, Jun 2019. [Contributed Talk]

- PowerPoint Slides
- Meeting Report

20) Davis, B. L. & Graham, A. W., "The Latest Black Hole Mass Scaling Relations," *Supermassive Black Holes: Environment and Evolution*, Jun 2019. [Contributed Talk]

- PowerPoint Slides

19) Davis, B. L., Graham, A. W., Sahu, N., & Cameron, E., "Black Hole Mass Scaling Relations," *American Astronomical Society Meeting Abstracts, Vol. 221, 206.02*, Jun 2019. [Contributed Talk]

- PowerPoint Slides

18) Davis, B. L., "Black Hole Mass Scaling Relations for Spiral Galaxies," *Astrophysics Colloquium: The University of Melbourne*, Sep 2018. [Seminar Talk]

- PowerPoint Slides

17) Davis, B. L., "Black Hole Mass Scaling Relations for Spiral Galaxies Determined from Pitch Angles and Multicomponent Structural Decompositions," *ASA 2018 Annual Scientific Meeting*, Jun 2018. [Poster Presentation]

- Video (starts at 1:25:55)
- Poster

16) Davis, B. L., "Black Hole Mass Scaling Relations for Spiral Galaxies Determined from Pitch Angles and Multicomponent Structural Decompositions," *Galactic Rings: Signposts of Secular Evolution in Disk Galaxies*, May 2018. [Contributed Talk]

- Video (starts at 3:49:30)
- PowerPoint Slides

15) Davis, B. L., "Black Hole Mass Scaling Relations for Spiral Galaxies," *California Institute of Technology, Astronomy Tea Talks*, Jan 2018. [Seminar Talk]

14) Davis, B. L., "Updating the (Supermassive Black Hole Mass) – (Spiral Arm Pitch Angle) Relation: A Strong Correlation for Galaxies with Pseudobulges," *National Astronomical Observatories, Chinese Academy of Sciences, Colloquium*, Oct 2017. [Seminar Talk]

13) Davis, B. L., "Updating the (Supermassive Black Hole Mass) – (Spiral Arm Pitch Angle) Relation: A Strong Correlation for Galaxies with Pseudobulges," *ASA 2017 Annual Scientific Meeting*, Jul 2017. [Contributed Talk]

12) Davis, B. L., "Spiral Arm Pitch Angle: Measurement and Relationship to Spiral Galaxy Structure and Supermassive Black Hole Mass," *ANITA Workshop 2017*, Feb 2017. [Contributed Talk]

11) Davis, B. L., "Spiral Arm Pitch Angle: Measurement and Relationship to Spiral Galaxy Structure and Supermassive Black Hole Mass," *Swinburne University of Technology, Centre for Astrophysics and Supercomputing, Colloquia Series*, Dec 2016. [Seminar Talk]

10) Davis, B. L., "A Fundamental Plane of Spiral Structure in Disk Galaxies," *A Lowell Observatory Workshop: The Formation and Evolution of Exponential Disks in Galaxies*, Oct 2014. [Poster Presentation]

9) Davis, B. L., "The Black Hole Mass Function Derived from Local Spiral Galaxies," *Friday Scientific Lunch Talk at NOAO-Tucson*, Nov 2013. [Seminar Talk]

8) Davis, B. L., Berrier, J. C., Johns, L., Shields, D. W., Kennefick, D., Kennefick, J., Seigar, M. S., & Lacy, C. H. S., "The Black Hole Mass Function Derived from Local Spiral Galaxies," *Structure and Dynamics of Disk Galaxies, ASP Conference Series, Vol. 480, 2014, p. 204*, Aug 2013. [Contributed Talk]

7) Davis, B. L., "Logarithmic Spiral Arm Pitch Angle of Late-Type Galaxies: Measurement and Relationship to Galactic Nuclear Supermassive Black Hole Mass," *American Astronomical Society Meeting Abstracts, Vol. 221, 206.02*, Jan 2013. [Contributed Talk]

6) Davis, B. L., Berrier, J. C., Johns, L., Shields, D. W., Kennefick, D., Kennefick, J., Seigar, M. S., Lacy, & C. H. S, "The Local Black Hole Mass Function Derived from Spiral Galaxies," *American Astronomical Society Meeting Abstracts, Vol. 220, 430.08*, Jun 2012. [Poster Presentation]

5) Davis, B. L., Berrier, J. C., Johns, L., Shields, D. W., Kennefick, D., Kennefick, J., Seigar, M. S., Lacy, & C. H. S, "The Local Black Hole Mass Function Derived from Spiral Galaxies," *42nd Annual Mid-American Regional Astrophysics Conference*, Apr 2012. [Contributed Talk]

4) Davis, B. L., Berrier, J. C., Shields, D. W., Kennefick, J., Kennefick, D., Seigar, M. S., Lacy, C. H. S., & Puerari, I., "Measurement of Galactic Logarithmic Spiral Arm Pitch Angle Using Two-Dimensional Fast Fourier Transform Decomposition," *American Astronomical Society Meeting Abstracts, Vol. 219, 246.01*, Jan 2012. [Poster Presentation]

3) Davis, B. L., "Measurement of Galactic Logarithmic Spiral Arm Pitch Angle Using Two-Dimensional Fast Fourier Transform Decomposition," *Annual Mid-American Regional Astrophysics Conference*, Apr 2011. [Contributed Talk]

2) Davis, B. L., Berlanga Medina, J. E., Shields, D. W., Kennefick, J., Kennefick, D., Berrier, J., Seigar, M. S., Lacy, C. H. S., & AGES, "Investigating the Clustering and Color of Galaxies in the COMBO-17 Chandra Deep Field South Survey and Possible Effects on Spiral Arm Pitch Angle," *Bulletin of the American Astronomical Society, Vol. 42, American Astronomical Society Meeting Abstracts #215, 435.17*, Jan 2010. [Poster Presentation]

1) Davis, B. L., Chevrier, V. F., Altheide, T. S., & Swaffar, C., "Reflectance Spectra of Low-Temperature Chloride and Perchlorate Hydrates and Their Relevance to the Martian Surface," *Lunar and Planetary Institute Science Conference Abstracts, Vol. 40, 1387*, Mar 2009. [Contributed Talk]

TEACHING EXPERIENCE

General Physics II Arkansas Tech University — Jan 2016 – May 2016
Visiting Assistant Professor of Physics

Introduction to Physical Science, Arkansas Tech University — Aug 2015 – May 2016
Visiting Assistant Professor of Physics

Physical Science Laboratory, Arkansas Tech University — Aug 2015 – May 2016
Visiting Assistant Professor of Physics

Physics Laboratory I, Arkansas Tech University — Aug 2015 – May 2016
Visiting Assistant Professor of Physics

Physics Laboratory II, Arkansas Tech University — Aug 2015 – May 2016
Visiting Assistant Professor of Physics

University Physics I, University of Arkansas — Aug 2015 – Dec 2015
Visiting Assistant Professor of Physics

Private Violin Lessons, taught from my home in Goshen, AR — 2013 – 2014
4 students, weekly lessons

Survey of the Universe, University of Arkansas — Aug 2013 – Dec 2014
Teaching Assistant

Survey of the Universe Laboratory, University of Arkansas — Aug 2012 – Aug 2013
Instructor

College Physics II Drill, University of Arkansas — Jan 2012 – May 2012
Instructor

University Physics I Laboratory, University of Arkansas — Aug 2008 – Dec 2008
Instructor

Astronomy Laboratory, Pittsburg State University — Jan 2006 – May 2008
Teaching Assistant

AWARDED OBSERVING TIME

Three Nights — Nov 2013
WIYN 3.5-m Telescope, pODI
For observation of AGN with spiral galaxy hosts at $z < 0$.

- News Article Coverage - University of Arkansas News: "Physicists Awarded Telescope Time at Kitt Peak National Observatory"

ACADEMIC HONORS & AWARDS

2nd Place in the Physics Division of the "Abstract to Contract" Fifth Annual Graduate Student Research Symposium — Feb 2012
University of Arkansas Graduate School
For placing in a poster presentation competition amongst other physics graduate students at the University of Arkansas.

- News Article Coverage - University of Arkansas News: "Winners Announced for From Abstract to Contract: Graduate Student Research Competition"

1st Place in the Physical Science Division of the "Abstract to Contract" Third Annual Graduate Student Research Symposium — Feb 2010
University of Arkansas Graduate School
For winning a poster presentation competition amongst other physical science graduate students at the University of Arkansas.

Honorable Mention for the Chambliss Astronomy Achievement Student Award — Jan 2010
215th American Astronomical Society (AAS) Meeting
For recognition of exemplary research by undergraduate and graduate students who present at one of the poster sessions at the meetings of the AAS.

Mendenhall Top Senior Math Student — May 2008
Pittsburg State University Department of Mathematics
For having the highest GPA among the graduating mathematics seniors.

Kansas Governor's Scholar — May 2003
Kansas State Department of Education
The Governor's Scholars program honors the top academic one percent of Kansas high school seniors.

Kansas Honor Scholar — Oct 2002
University of Kansas Alumni Association
A distinction for high school seniors who demonstrate a consistent record of superior academic achievement merits recognition as outstanding in the State of Kansas.

FELLOWSHIPS, SCHOLARSHIPS, & GRANTS

International Astronomical Union (IAU) Symposia Travel Grant — Nov 2019
IAU Symposium 359: Galaxy Evolution and Feedback Across Different Environments (GALFEED)
€1000 travel grant to attend the GALFEED conference at Bento Gonçalves, RS, Brazil, March 2-6, 2020.

Research Infrastructure Program — 2014
Arkansas Space Grant Consortium
Project: The Host Galaxies of Active Galactic Nuclei at Moderate Redshifts

Graduate Fellowship Program — 2013
Arkansas Space Grant Consortium

Scholarship Program — 2011
Arkansas Space Grant Consortium
Project: An Investigation of the Evolution of Spiral Galaxies Using N-Body Simulations and the Possible Effects of Dark Matter Concentration and Galaxy Environment.

Doctoral Academy Fellowship — 2008 – 2012
University of Arkansas Graduate School
Supplemental fellowship from the graduate school to the top ≈40 incoming doctoral students as determined by GRE/GMAT scores.

CAMPUS ACTIVITIES

University Symphony Orchestra, University of Arkansas — Aug 2009 – May 2016
Violin (Section I, Concertmaster, Section II Principal), Viola (Principal)

Southeast Kansas Symphony Orchestra, Pittsburg State University — Aug 2003 – May 2008
Violin (Section I, Concertmaster, Section II), Viola (Principal)

The PSU Chamber Orchestra, Pittsburg State University — Aug 2003 – May 2008
Violin (Section I, Concertmaster, Section II), Viola (Principal)

OTHER WORK EXPERIENCE

Arkansas Philharmonic Orchestra, Bentonville, Arkansas, USA — May 2009 – May 2016
Section Violin I

Litigation Resource Group, Fort Smith, Arkansas, USA — Jul 2012 – Feb 2013
Videographer/Accident Reconstruction Consultant

North Arkansas Symphony Orchestra, Fayetteville, Arkansas, USA — Nov 2006 – Apr 2008
Section Violin I & II

Sentinel Times, Galena, Kansas, USA — Jun 2000 – Jun 2008
Junior Editor/Mail & Delivery Services

Old Riverton Store, Riverton, Kansas, USA — Jun 1998 – Jun 2008
Clerk/Salesman/Food Service/Horticulturist

Freelance Musician Kansas, Missouri, & Arkansas — Jun 1996 – Jul 2016
Violinist/Violist

LANGUAGES

English: Native language

CITIZENSHIP

US Citizen; Born in Joplin, Missouri, USA

SKILLS, INTERESTS, & EXPERTISE

Typing Proficiency: 67 WPM (285 CPM) with 100% accuracy

TEX, LATEX, XƎTEX, MATLAB, Mathematica, R, Adobe Creative Cloud, IRAF, DS9, Gimp, Supermongo, Microsoft Office Suite, Unix/Linux, Fortran, C++, IDL, HTML, Google Analytics, Python, MPI, Machine Learning, SQL, HPC, Problem Solving, Data Science, Data Engineering, Data Analysis, Observing, Classical Music, Music Performance, Oration, Teaching, Logarithmic Spiral Arm Pitch Angle, Multicomponent Galaxy Decomposition, Telescopes, Black Holes, Data Reduction, Galaxy Evolution, Astronomy, Astrophysics, Active Galactic Nuclei, Galaxy Structure, Galaxy Dynamics, Quasars, Galaxy Clusters, Dark Matter, Galaxy Formation, Space & Planetary Sciences, Martian Surface Chemistry, Golf, Tennis, Baseball, Basketball, Violin, Viola, Guitar, Bass, Piano, Ukulele, Saxophone, Singing, SCUBA Diving, Carpentry, Woodworking, Games (Card, Board, & Video), etc.

REFERENCES

Dr. Andrea Valerio Macciò (Postdoctoral Supervisor)
Associate Professor and Program Head of Physics
New York University Abu Dhabi, Center for Astro, Particle, and Planetary Physics
PO Box 129188, Saadiyat Island, Abu Dhabi, United Arab Emirates
maccio@nyu.edu • +971 2 6284386

Dr. Alister Graham (Postdoctoral Supervisor)
Professor
Swinburne University of Technology, Centre for Astrophysics and Supercomputing
Mail Number H29, PO Box 218, Hawthorn, Victoria 3122, Australia
agraham@astro.swin.edu.au • +61 3 9214 8784

Dr. Julia Kennefick (Ph.D. Advisor)
Associate Professor
University of Arkansas, Department of Physics
825 West Dickson Street
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